

PFNs learn structured spectral representations that are linearly decodable, causally active, and extractable as explicit GP kernels in a single forward pass.



Mechanistic Evidence for Spectral Structures in Prior-Data Fitted Networks

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Motivation & Setup

Prior-Data Fitted Networks (PFNs):

The Promise: Instant predictions on new tasks without per-task training.

The Problem: Structural opacity. It is unknown if they learn structured prior assumptions or merely memorize training mappings.

Core Questions:

- Do PFNs encode spectral information in a structured form that is causal?
- Can we extract this structure as explicit, portable GP kernels?

Architectures studied: VA-PFN (joint attention), DVA-PFN (decoupled attention), and TabPFN (alternating attention).

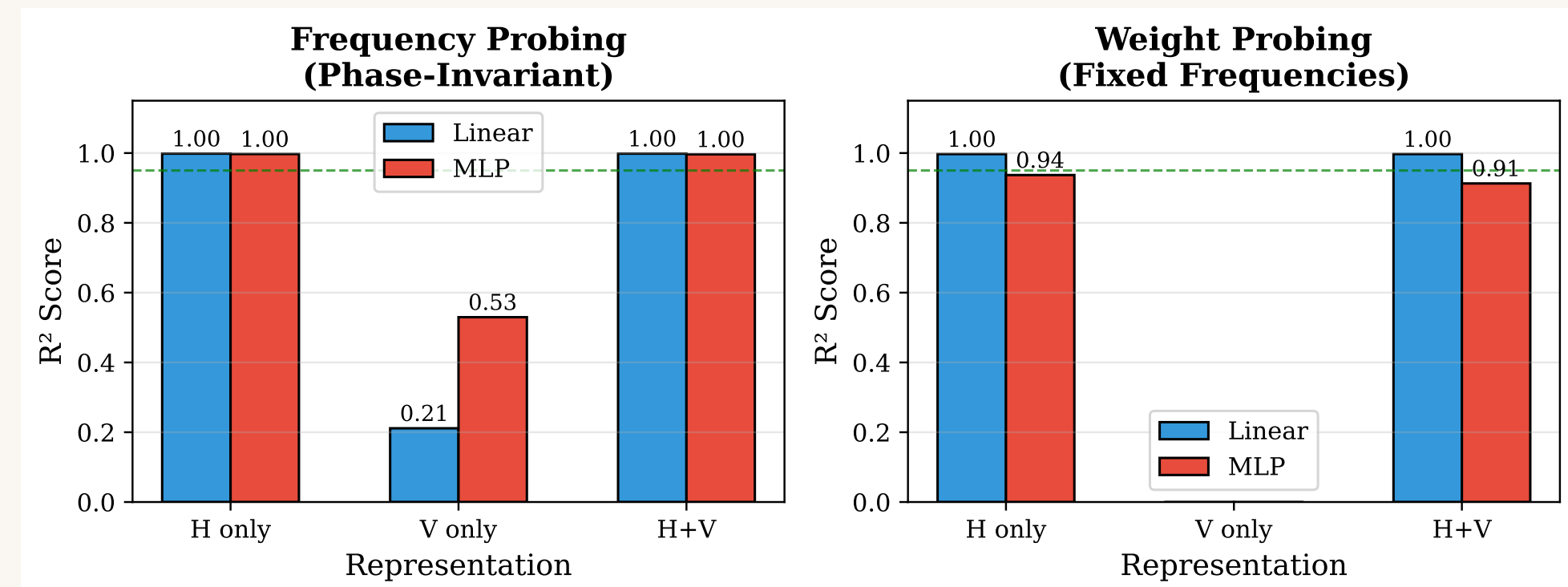
Probing: Linearly Decodable Structure

We probe latent attention representations using linear and nonlinear (MLP) models:

Linear Decodability: The performance gap between linear and MLP probes is $\leq 2\%$ everywhere, indicating that spectral properties are represented linearly.

Generality: Spectral structure also emerges on 5D GP inputs (RBF: $\rho_\Delta = 0.82$; Matérn: $\rho_\Delta = 0.86$), not just 1D sinusoids.

Key finding: Linear probes recover generating frequency with $R^2 \geq 0.967$ across all architectures. On TabPFN (not trained on spectral data): $R^2 = 0.993$.



Probing R^2 scores for generating frequency and mixture weight across architectures.

Causal Evidence: Activation & Subspace Patching

Probing confirms information is *present*, but not necessarily *causally used*. We close this gap using causal interventions.

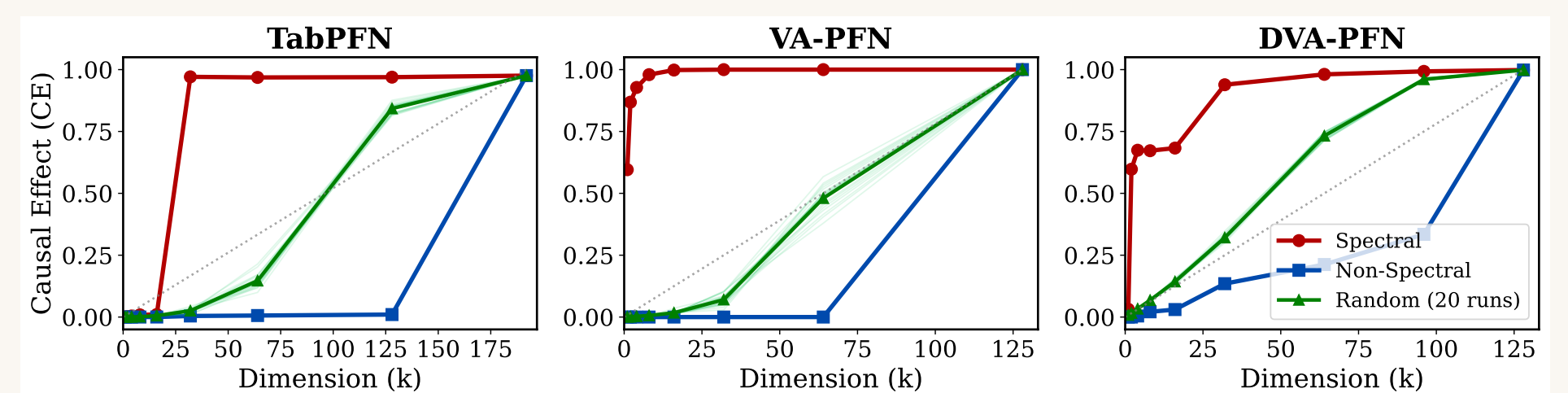
1. Activation Patching: Total Causal Effect

Swapping attention representations $\tilde{H}_A^{(\ell)} \leftarrow \tilde{H}_B^{(\ell)}$ measures the total Causal Effect:

- Result:** **CE = 0.999** at layer $\ell \geq 2$ ($p < 10^{-133}$).
- Dynamics:** A sharp $L_1 \rightarrow L_2$ transition shows that spectral structure emerges in a **single cross-attention step**.

2. Subspace Patching: Low-Dimensional Selectivity

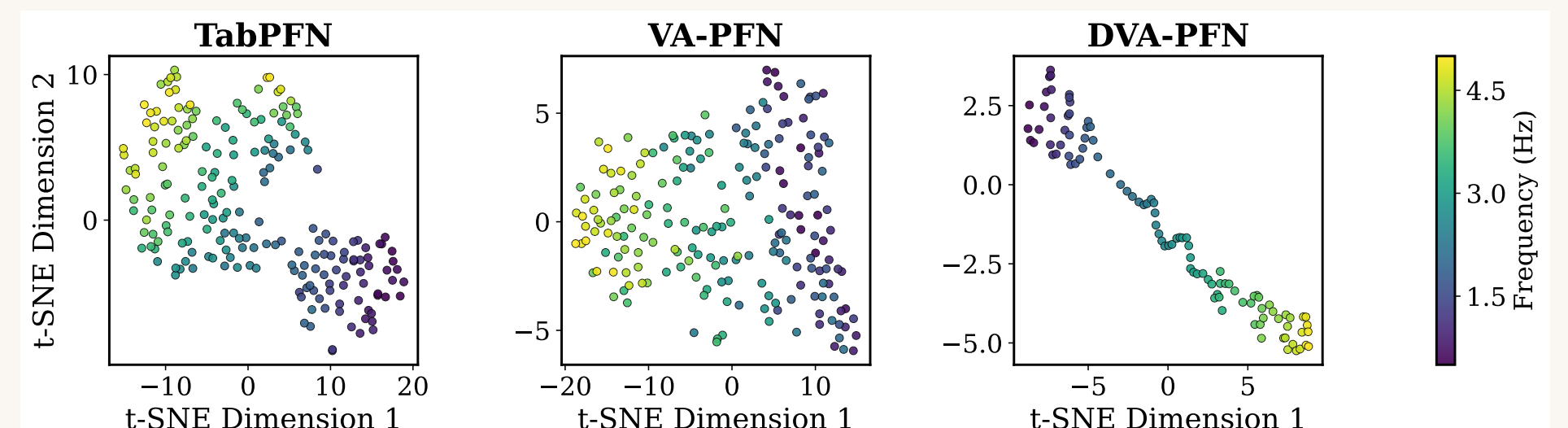
We patch only the top- k PCA directions of $\tilde{H}$ ranked by frequency correlation:



Dose-response curves. Spectral PCs (red) transfer identity at $10\text{--}34\times$ the rate of random directions (green).

Patching only **2 PCs** (1.5% of dimensions) transfers **59.2%** of spectral identity with **33.9x** selectivity.

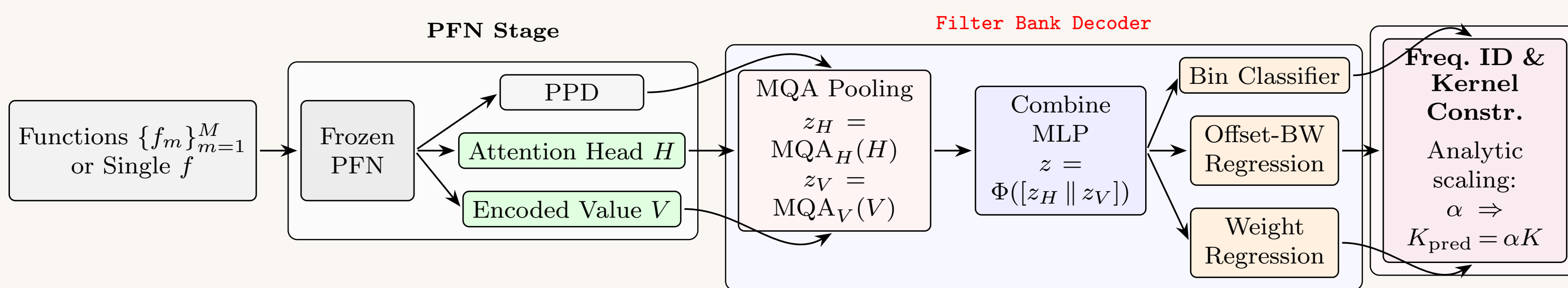
Attention Manifold Structure Across Architectures



t-SNE of mean-pooled representations $\tilde{H}$, colored by frequency, visualizations for spectral structure.

Filter Bank Decoder: Extracting Portable Kernels

We introduce the **Filter Bank Decoder** — a lightweight read-out mapping PFN latents to explicit spectral densities via Bochner's theorem: $k(\tau) = \sum_{q=1}^Q w_q \exp(-2\pi^2 \sigma_q^2 \tau^2) \cos(2\pi \mu_q \tau)$



Zero test-time optimization.

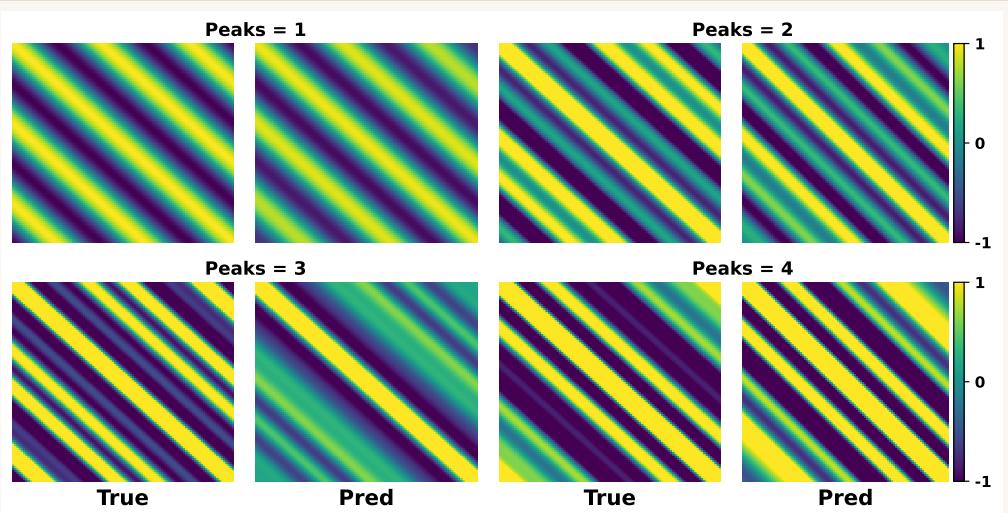
Single forward pass produces a portable GP kernel supporting downstream tasks (e.g. CPU-only BO) without network re-evaluation.

Architecture:

Pools attention streams, predicts bin activation probabilities, offsets, bandwidths, and weights.

Decoder Results: Baselines & Kernel Recovery

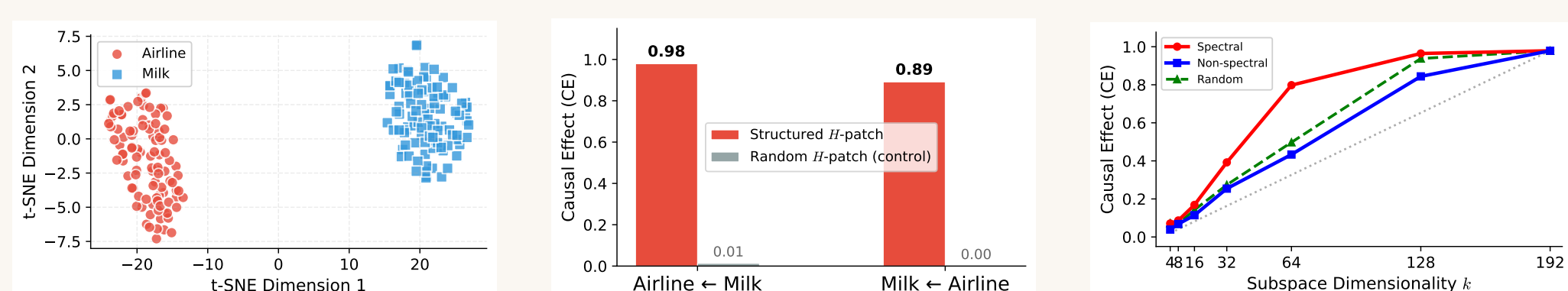
Kernel	Ours	DKL	RFF
RBF	1.1×10^{-3}	8.4×10^{-4}	1.0×10^{-3}
Periodic	1.5×10^{-3}	6.6×10^{-3}	6.4×10^{-3}
RBFxPer	2.7×10^{-3}	4.9×10^{-3}	4.6×10^{-3}
SM ($Q=4$)	1.5×10^{-3}	1.1×10^{-3}	1.4×10^{-3}
Time (s)	0.004	0.918	0.758



Left: GP regression MSE on cookbook benchmarks ($\sim 250\times$ speedup). Right: True vs. decoded kernels.

Real-World Validation of Mechanistic Insights

Airline Passengers and Milk Production, converted to 5D tabular regression.



Left: t-SNE separates series. Mid: CE = 0.98. Right: Spectral subspace is $2\times$ more efficient.

Conclusions & Limitations

Conclusions:

PFNs encode spectral structures that are **linearly decodable**, concentrated in a low-dimensional causal subspace, and emergent across architectures.

The **Filter Bank Decoder** extracts explicit, portable GP kernels in a single forward pass, enabling CPU-only Bayesian optimization.

Limitations:

Restricted to stationary spectral-mixture kernels; multi-dimensional settings under investigation.

Decoder is diagnostic; a small predictive MSE gap remains relative to full PFN representation.



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